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Prof. Kenneth M. Merz, Jr., Editor-in-Chief

Journal of Chemical Information and Modeling, American Chemical Society

Dear Prof. Merz,

We submit our manuscript entitled **“Estimand Divergence Between Static Docking and Molecular Dynamics as a Triage Filter for Antimalarial Leads”** for consideration as a research Article in the *Journal of Chemical Information and Modeling*.

Overcoming multi-target drug resistance in *Plasmodium falciparum* is a central goal in antimalarial lead discovery. Static docking scores capture unrelaxed grid complementarity, ignoring dynamic pocket relaxation upon mutation. We ask whether static grid docking and short explicit-solvent molecular dynamics estimate the same target quantity across a resistance-mutant panel, and whether their directional disagreement can flag states for prospective biochemical testing. We apply both procedures to 396 African natural products and 454 synthetic antimalarials across 136 wild-type and mutant structures of PfDHFR and PfCRT.

We define a Resistance-Resilience Score (RRS) for score retention across mutant panels, excluding non-binding wild-type baselines ($|S_{WT}| < 5.0 \text{ kcal mol}^{-1}$). In short explicit-solvent MD (10 ns), static scores and trajectory pose retention diverged directionally in 87.5 % (7/8; 95% Wilson CI 52.9 %–97.8 %, $n = 8$) of matched comparisons — distinct physical estimands (unrelaxed grid complementarity vs. local dynamic accommodation), not docking failure. Per Soares et al. (2023), these short trajectories are secondary structural checks: a calibration signal, not a validated prediction.

The observation is bounded by three stress tests, two negative and reported as first-class findings: (i) Vina enrichment on DEKOIS 2.0 PfDHFR was at chance level (ROC-AUC 0.450, $\text{EF}_{5\%} = 0.00$); (ii) rigid $\Delta\Delta G$ did not track curated ChEMBL genotype-resolved fold-shifts (28 paired compounds; $\rho -0.13$ – -0.19 , agreement 28.6 %–41.7 %); (iii) RRS, PNS, and ACSI were mutually non-associated ($\rho \in [-0.41, -0.21]$, adjusted $p > 0.50$) and are reported as separate dimensions. To our knowledge, contemporaneous docking–MD antimalarial studies report neither a concordance quantification nor comparable negative controls. A claim-ledger table and practical-guidance paragraph assign each central claim its evidential status.

The main contributions of this work include:

1. *Estimand divergence* between rigid docking retention and dynamic pocket relaxation as a protocol-local triage observation (7/8 matched states, single 10 ns replicate scope).
2. Protocol-stability check on a 39-compound panel (312 complexes) with class-level agreement under GNINA CNN rescoring (no rank transfer claimed).
3. An explicit claim ledger with evidential status per claim — including two negative controls — plus four operational triage rules.

We believe this calibration-forward treatment of docking–MD disagreement falls squarely within the methodological scope of *JCIM*.

All code, configuration files, and datasets are publicly archived on Zenodo at DOI: 10.5281/zenodo.22829031. This manuscript is original, has not been submitted elsewhere, and has been approved by all co-authors.

Suggested Reviewers: Dr. Fidele Ntie-Kang (Univ. of Buea, fidele.ntie-kang@ubuea.cm) — NP chemoinformatics; Dr. Kelly Chibale (Univ. of Cape Town, kelly.chibale@uct.ac.za) — antimalarial chemistry; Dr. John D. Chodera (MSKCC, john.chodera@choderalab.org) — free-energy/MD methods.

Sincerely,

Myke Vital Sao Temgoua

(on behalf of all co-authors)